

# TARS

## Sovereign Company Brain for Early-Stage Startups

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- Track chosen: Sovereign AI



## Four fatal frictions that stall early-stage engineering teams.

### 01 Rapid Pivots & Toxic Context

HIGH

**Problem** Startups pivot business models violently — chats, PRs, and docs stay full of obsolete assumptions.

**Consequence** New hires and AI agents inherit “dead context,” building features for defunct models.

### 02 Bus Factor = 1 Amnesia

CRITICAL

**Problem** In 4-person teams, one engineer builds auth or billing in a 72-hour sprint with zero documentation.

**Consequence** When that engineer leaves, 100% of subsystem institutional memory vanishes, freezing velocity.

### 03 Founder Interruption Tax

ACUTE

**Problem** The technical founder is the sole routing switch holding architecture and sales promises in their head.

**Consequence** Junior hires ping them 15x/day, slashing founder output 50% while the runway clock ticks.

### 04 Disconnected Client Promises

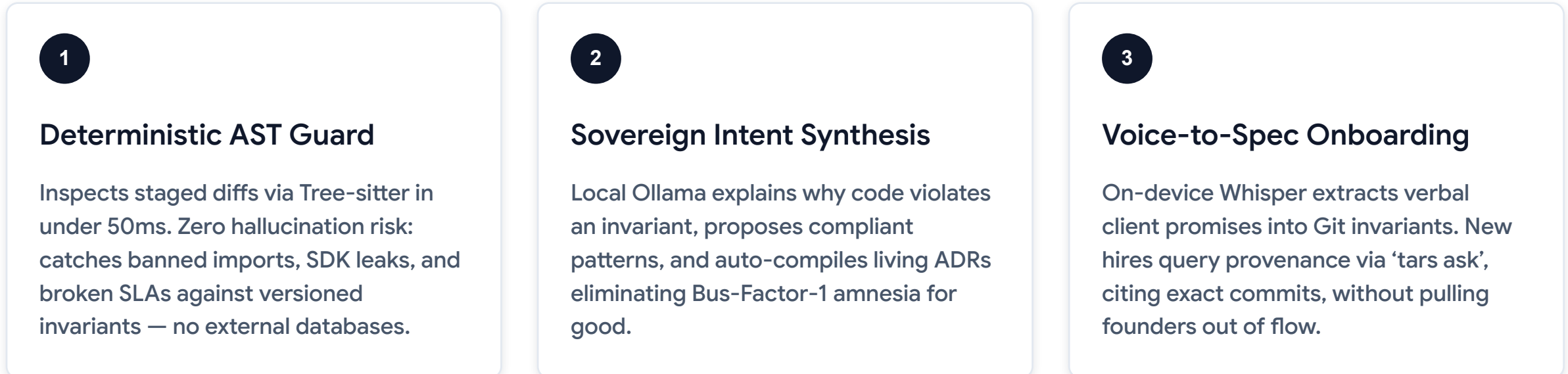
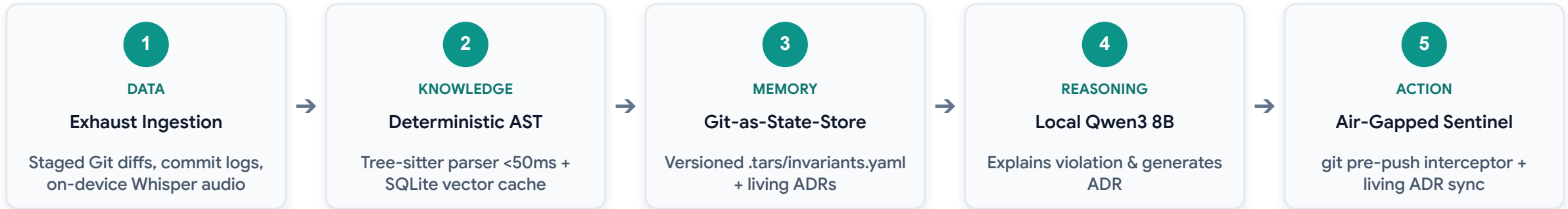
HIGH

**Problem** Crucial commitments made on customer calls evaporate before engineering sprints are planned.

**Consequence** Broken enterprise promises kill contracts — and strict client NDAs forbid cloud AI tools.



## TARS operationalises the 5-step sovereign pipeline end-to-end, on-device.





# ASYNC 2026



Generic tools trade sovereignty for convenience. TARS refuses that trade-off.

| CAPABILITY         | GENERIC RAG                | STATIC WIKIS             | CLOUD AI (COPILOT)       | TARS                          |
|--------------------|----------------------------|--------------------------|--------------------------|-------------------------------|
| Data Sovereignty   | ✗ Multi-tenant cloud leaks | ✗ Centralized SaaS       | ✗ Streams code to cloud  | ✓ 100% air-gapped, 0.00 KB    |
| Org. Memory        | ✗ Stateless sessions       | ⚠ Unmaintained docs      | ✗ Editor tab only        | ✓ Git-as-State-Store (.tars)  |
| Knowledge Model    | ✗ Flat vector chunks       | ✗ Unlinked folders       | ⚠ Text autocomplete      | ✓ Deterministic AST + vectors |
| Operational Action | ✗ Passive Q&A chatbot      | ✗ Passive text graveyard | ⚠ Inline completion only | ✓ Native pre-push sentinel    |
| Team Alignment     | ✗ Single-user tool         | ⚠ Manual maintenance     | ✗ Individual dev tool    | ✓ Voice-to-spec + team CLI    |

**The Unfair Startup Moat** — Enterprise tools like Glean or Confluence fail in startups: they demand dedicated technical writers and multi-year budgets. TARS thrives on startup chaos — zero-effort exhaust capture, deterministic AST guards, and 100% air-gapped protection of proprietary IP.



## Feasibility & Execution

### 1 The Sovereign Tech Stack

**CORE ENGINE & PARSER** Python 3.11 + Tree-sitter

**Local LLM inference** Ollama running Qwen3 8B

**Sovereign Storage & CLI** Kuzu embedded graph database + SQLite metadata. Zero daemon, packaged as a single native CLI binary (`tars`).

**Local Dashboard** React + Tailwind CSS

**Audio** Whisper.cpp / pywhispercpp, on-device

✓ Zero Docker · Zero daemons · 100% on-device

### 2 MVP Scope

**AST Parser:** Tree-sitter diff checks in < 150ms.

**Rule Engine:** Layer boundaries via .tars/invariants.yaml.

**Local Qwen3 8B:** Air-gapped reasoning; zero cloud egress.

**Living ADRs:** Auto-drafts Markdown records on block.

**Native CLI:** Lightweight, zero-daemon git pre-commit hook.

✓ Gate: Offline CLI echo test passing

### 3 Deliverables

**Scenario** The Airplane-Mode Demo — 100% air-gapped, live

1. **Air-Gap** Laptop Wifi switched OFF; system fully operational

2. **Voice-to-Spec** Client audio transcribed on-device in 3.5s

3. **AST Intercept** Banned SDK commit blocked by pre-push in 28ms

4. **Rationale** Local Ollama explains violation, 0.00 KB egress

🔒 Gate: Live push intercepted, 0.00 KB egress



# ASync 2026



## Three quantifiable outcomes, proven live with a physically air-gapped stage demo.

### 80% FASTER

Developer Onboarding Time

Cuts time-to-first-commit from 3 weeks to 3 days via automated provenance search.

### 15+ HRS / WK

Founder Deep-Work Preserved

Shields technical founders from repetitive pings, protecting cash runway.

### 100% PRIVATE

Zero Cloud Egress Guaranteed

Airplane-Mode proof: code, financials, and client audio never leave the laptop.

#### 1 Expected Real-World Impact

- Runway Preservation: eliminates \$20k+ wasted on dead-end rewrites and contractor black-boxes.
- Eliminates Bus Factor = 1: living ADRs mean an early departure never paralyzes velocity.
- Democratizes Compliance: lets 3-person pre-seed teams enforce strict client SLAs from day one.
- Zero Infrastructure: no Docker sprawl or external servers — state lives in the Git repo.

#### 2 Demo Outcome

- The Airplane-Mode : full closed loop executed live with Wi-Fi disabled.
- Deterministic + LLM Hybrid: sub-50ms AST detection paired with deep Ollama explanation.
- Empirical Zero-Egress Proof: physical disconnection proves true sovereignty, on stage.
- Pragmatic Startup Fit: Git-as-State-Store, zero maintenance burden, ready day one.